

MUHAMMAD FAKHRI HELMI

PETROLEUM ENGINEER | DRILLING & WELL OPERATIONS | WELL INTEGRITY | FIELD SUPPORT

Jakarta, Indonesia | +62 852-6150-5740 | muhammadfakhrihelmi@gmail.com | [linkedin.com/in/fakhri-helmi](https://www.linkedin.com/in/fakhri-helmi)

PROFESSIONAL PROFILE

Petroleum Engineering graduate with more than one year of upstream oil and gas experience through the Asia Serv Future Leader Program, supporting PHE OSES well-integrity operations and the INPEX Masela technical study. Coordinates field crews, equipment readiness, HSE documentation, well data, client reporting and multidisciplinary study interfaces; also built a Well Integrity Management System and interactive 3D technical visualizations. Earlier drilling-engineering internship experience covers Daily Drilling Reports, drilling time and NPT, parameters, BHA and downhole tools. Seeking junior opportunities in drilling and well operations, well services/integrity, field engineering, technical support or project coordination; open to field deployment, rotation and relocation.

CORE COMPETENCIES

Drilling & Well Planning • Daily Drilling Report and NPT Review • Well Integrity & Barrier Awareness • Wellhead/X-mas Tree Maintenance • Field Crew, Equipment & Mobilization Readiness • HSE Plan, PTW, JSA & Risk Review • Technical Data Analysis & Visualization • Client Reporting & Project Coordination

PROFESSIONAL EXPERIENCE

PT ASIA SERV INDONESIA — Asia Serv Future Leader Program / Project & Technical Coordinator Jul 2025 – Sep 2026

PHE OSES Project — Well Integrity, Wellhead & X-mas Tree Maintenance

- Coordinated two vessel-based field-service crews of four personnel each, supporting more than 1,100 wellhead/X-mas tree inspection, testing and maintenance visits; delivered at least 84 visits per month with a peak of 120.
- Managed execution readiness across crew mobilization, equipment and certification schedules, HSE plans, PTW/JSA support and client coordination; reviewed approximately 5–6 field reports per operating day and followed up technical or documentation gaps.
- Built a Well Integrity Management System using Node.js and Python-assisted workflows, consolidating 7,713 valve-inspection and 1,418 troubleshooting records with visit history, pressure mapping, integrity status and plan-versus-actual tracking.
- Supported troubleshooting records across approximately 735 unique wells, including X-tree valve repair, annulus pressure testing using a Haskel pump, and wellhead seal/cavity checks; enabled same-day review and reduced duplicate data handling.

Digital project evidence: [View WIMS Project Coordinator Portfolio](#)

- Built an Excel/VBA invoice-summary workflow and introduced earlier document-signature controls, helping reduce the typical invoice-to-payment cycle from 3–4 months to approximately 1.5 months.
- Maintained field-delivery quality reflected in a verified client satisfaction score of 4.20/5 across nine signed PHE OSES evaluations, with recurring positive feedback on responsiveness and coordination.

INPEX Masela Project — Conductor & Completion Riser Technical Analysis Support

- Reviewed and interpreted technical-study outputs covering well loads, soil interaction, axial movement, offsets, wave-induced fatigue and vortex-induced vibration, then developed a browser-based 3D visualization to translate the analysis into a clear technical narrative for project review and presentation.
- Coordinated study inputs, scope and P&L follow-up, technical queries, closure of 26 client comments and issuance of a 94-page conductor-analysis report across INPEX, Asia Serv and the specialist consultant.

Technical project evidence: [View INPEX Masela Conductor Analysis 3D Storyboard](#)

PT PERTAMINA DRILLING SERVICES INDONESIA — Drilling Engineer Intern

Jul 2023 – Sep 2023

- Reviewed Daily Drilling Reports and analyzed drilling time, NPT, drilling parameters, BHA/downhole tools and operational events to support concise engineering insights for drilling-performance discussions.
- Applied Halliburton Landmark WellPlan concepts to evaluate drillstring and BHA behavior, strengthening practical understanding of well planning, drilling mechanics and field operations.

EDUCATION

UNIVERSITAS PERTAMINA — B.Eng. / S1 Petroleum Engineering

Sep 2020 – Aug 2024

GPA: 3.31/4.00 | Jakarta, Indonesia

Selected Academic and Technical Projects

Drillstring Design and Mechanical Load Analysis — Undergraduate Thesis

2024

- Compared three drillstring designs for the final 8½-in. section of an anonymized J-type directional well using Landmark WellPlan and manual checks for torque, drag, tension, stress and sinusoidal/helical buckling.
- Recommended Design A by optimizing component and HWDP selection without changing total BHA length; simulation-to-manual MAPE was approximately 8.5% for torque and 27% for tension.

Integrated Field Development Plan and Petroleum Economics — Capstone Project

2024

- Developed production forecasts and project economics covering CAPEX/OPEX, NPV, IRR, payback and a three-well break-even case; the modeled company economics favored PSC Gross Split over Cost Recovery under the project assumptions.

Interactive case study: [View Simprug Field Development 3D Economic Case Study](#)

ADDITIONAL PROJECT LEADERSHIP

- Supported three upstream tender opportunities covering well integrity and technical studies, with two awarded; coordinated readiness evaluation for 12 personnel and 28 primary/backup equipment categories within three weeks.
- Prepared 10 first-time OSES personnel through contract/JSA briefings, competency checks, equipment familiarization and field-readiness training; all mobilized and none were rejected by the client.
- Coordinated the 2026 OSES contract transition for eight personnel across two vessel crews, including HSE Plan, bridging, PTW and mobilization documentation, enabling operational continuity without crew demobilization.

PROFESSIONAL AFFILIATIONS AND ORGANIZATIONAL EXPERIENCE

- Jakarta Drilling Society — Active Member (May 2026–Present): participates in industry knowledge-sharing activities and the Alpha-05 Integrated Drilling Project; developed a browser-based 3D well-construction animation covering trajectory, casing, BHA, hydraulics, anti-collision, torque-and-drag considerations and drilling risks.

JDS project evidence: [View Alpha-05 Drilling 3D Animation](#)

- HMTM SPRUDA, Universitas Pertamina — Head of Quality Control (Jan 2023–Jul 2023): led organizational-governance reviews, evaluated divisional work programs and facilitated progress meetings to strengthen compliance and cross-division coordination.
- HMTM SPRUDA — Staff, Quality Control (Jul 2022–Jan 2023) and Cadre Division (Jul 2021–May 2022): reviewed work programs, supported organizational administration and helped design leadership-regeneration activities for petroleum-engineering students.
- LAKSAMANA II — Project Manager (Jan 2022–Feb 2022): planned and coordinated a student leadership program across multiple committees, including task delegation, readiness monitoring and event execution.
- LAKSAMANA I — Head Liaison Officer (Oct 2021–Mar 2022) and PROTECT 2022 — Operational Staff (Feb 2022–May 2022): supported a leadership and orientation program for 110 new students and coordinated operational resources during program delivery.

TECHNICAL SKILLS

Engineering: Halliburton Landmark WellPlan/EDM, Compass, StressCheck, IPM-PROSPER, CMG; torque and drag, directional-well mechanics, nodal-analysis fundamentals and reservoir-simulation exposure. **Digital & Business:** Advanced Excel/VBA/macros, Power BI, Python fundamentals, JavaScript/Node.js, Three.js, Microsoft Office, technical dashboards, project P&L and reporting.

SELECTED TRAINING AND CERTIFICATIONS

IDTC BOP Level 1 • Directional Drilling Essentials • Accounting & Cost Management for Oil & Gas • General OHS Basic • Advanced Excel

LANGUAGES AND MOBILITY

Bahasa Indonesia (Native) • English (Professional Working Proficiency) • Open to field/site work, rotation, shifts, relocation and deployment across Indonesia